CSCI 4760 : Spring 2026

Assignment 3: Implement Traceroute (10 points)

Overview

In this assignment, you will implement a simplified version of the Linux `traceroute` tool using Python.

Requirements

Your program should send **UDP probes** with increasing TTL (Time-To-Live) values to discover the path packets take to a destination. Specifically,

- Accept **one command-line argument** (destination hostname or IP)
- Send **UDP packets**
- Increment TTL from **1 to 20**
- Send **one probe per hop**
- Record RTT
- Use:
  - Random destination port in range **33434–33464**
  - Random high source port
- Print output similar to Linux traceroute:

Example:

```
1  router.local  (192.168.1.1) 2.454ms
2  *
3  example.net (10.0.0.1) 17.3434ms
```

- Print `*` if no response is received
- Stop when destination is reached

Packet Capture Requirement

You must use **Wireshark (or similar tool)** to capture traffic generated by your program.

Submit screenshots showing:

- Output of the your program when input is i) a domain and ii) an IP address
- Outgoing probe packets (with TTL highlighted)
- At least two different types of responses observed during execution

For each screenshot, include a short explanation (1–2 sentences) describing what is happening.

Questions (Answer briefly)

1. What is the purpose of the TTL field in traceroute? Explain whether and how this value changes as the packet moves through the network.
2. Why does traceroute use high UDP port numbers?
3. What do the different responses you observed indicate?
4. If there is no response at hop x (shown as *), will all subsequent hops also show *?
5. Briefly describe how traceroute operates in Windows.

Deliverables

Submit a Zip file containing the following to eLC:

- Python file
- A PDF document that includes answers to the questions and captured screenshots.
